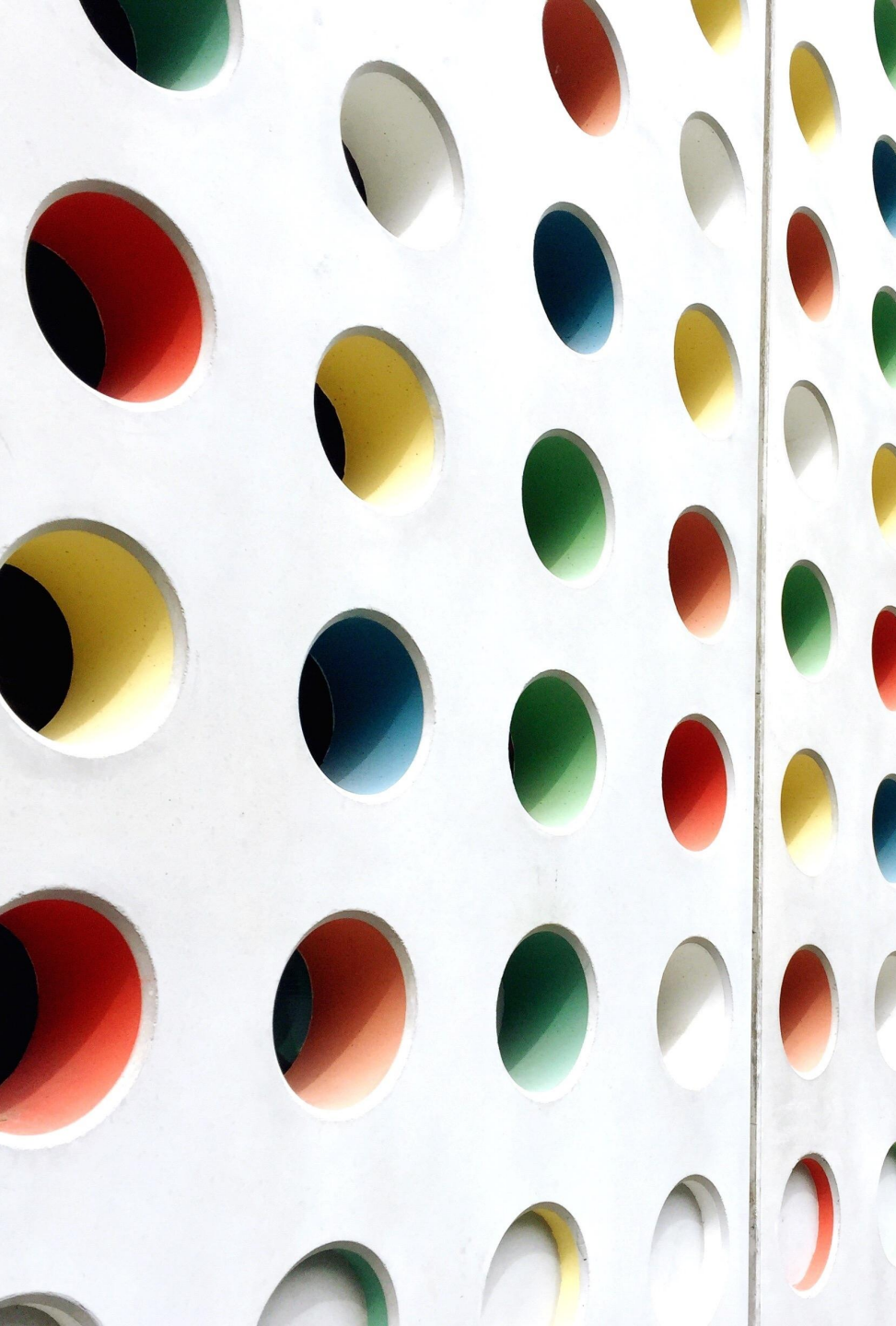




XML Visualization, XPath, XSLT

BY

SANA RIZWAN

Visualize XML

- Paste your XML into <https://codebeautify.org/xmlviewer> to view it as a tree.
- This will help to design an Xpath.



- Nodes: something you can select.
 - Elements, attributes, text, comment, namespace.
- Relationships between nodes are self explanatory:
 - Parent, child, sibling, ancestor, etc.

Demonstration

<https://codebeautify.org/xmlviewer>

Free Online XPath Tester

Best Online XML Viewer

Securehttps://codebeautify.org/xmlviewer

14F54045 - ScrumpyCustomize LinksShare on ConfluenceFree HotmailShare on ConfluenceWindows MarketplaceWindows MediaWindowsUCitLabsOther bookmarks

Code BeautifyJSON Formatter | My Ip | Search | Recent Links | Sample | More | Sign in | (?)

XML VIEWER

Save & Share

XML Input

sample

```
3  <?xml version="1.0" encoding="UTF-8" ?>
4  <xsi:noNamespaceSchemaLocation="plants.xsd" ?>
5  <!-- All payload data belongs here -->
6  <genus>Cercis</genus>
7  <species>canadensis</species>
8  <common_name>Eastern Redbud</common_name>
9  <height>25</height>
10 <begin_bloom_date>2017-03-15</begin_bloom_date>
11 <end_bloom_date>2017-03-30</end_bloom_date>
12 <edible>true</edible>
13 <sun_tolerance>full_sun</sun_tolerance>
14 <bloom_color>violet</bloom_color>
15 <specimens>
16   <specimen>
17     <latitude>39.47</latitude>
18     <longitude>-84.51</longitude>
19     <planted_by>Brandon Jones</planted_by>
20     <comments>Beautiful tree!</comments>
21   </specimen>
22   <specimen>
23     <latitude>52.40</latitude>
24     <longitude>1.00</longitude>
25     <planted_by>Joe Bloggs</planted_by>
26     <comments>RBG Kew</comments>
27   </specimen>
28   <specimen>
29     <latitude>-22.50</latitude>
30     <longitude>-68.25</longitude>
31     <planted_by>Guillermo Sanchez</planted_by>
32     <comments>Lovely Specimen!</comments>
33   </specimen>
34 </specimens>
```

Load Url

Browse

Tree View

Beautify / Format

Minify

Use Namecheap EasyWP at the Low Price of \$8.88

Get Started

XML to JSON

Result : XML Tree View

plant ..

@xmlns:xsi: http://www.w3.org/2001/XMLSchema-instance

@xsi:noNamespaceSchemaLocation: plants.xsd

genus Cercis

species canadensis

common_name Eastern Redbud

height 25

begin_bloom_date 2017-03-15

end_bloom_date 2017-03-30

edible true

sun_tolerance full_sun

bloom_color violet

specimens ..

specimen ..

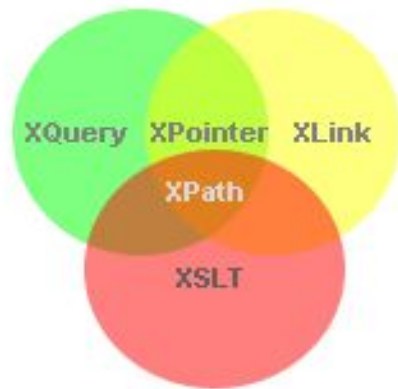
Why XPath

- Find what you want in an XML file.
- Implemented by many programming languages: Java, JavaScript, C#.
- Can easily test in a webpage:
<https://www.freeformatter.com/xpath-tester.html#ad-output>
- Used in XSLT!

What is XPath?

XPath is a major element in the XSLT standard.

XPath can be used to navigate through elements and attributes in an XML document.



- XPath is a syntax for defining parts of an XML document
- XPath uses path expressions to navigate in XML documents
- XPath contains a library of standard functions
- XPath is a major element in XSLT and in XQuery
- XPath is a W3C recommendation

XPath

XPath Terms

- Slash: / to navigate through a path on a tree.
 - Start at root. /plants/specimens, for example.
- Square brackets: [] to select a single item.
 - Number: [1]
 - Comparison: [latitude > 0]
 - Function: [last()]

- Use in the []
 - Math: +, -, *, div
 - Comparison: =, !=, <, <=, >, >=
 - Combination: and, or mod

- //: find an element anywhere.
 - //genus to find genus, no matter where it is.
- @: selects attributes. @class="top", for instance.
- . Current node.
- .. Parent of current node.

- /plant/specimens
- The first specimen:
 - /plant/specimens/specimen[1]
- The last specimen:
 - /plant/specimens/specimen[last()]
- Specimens with longitude greater than 3:
 - /plant/specimens/specimen[longitude>3]
- Comments from specimen with longitude greater than 3

Some Examples

Demonstration --

<https://www.freeformatter.com/xpath-tester.html>

Formatters

JSON Formatter
HTML Formatter
XML Formatter
SQL Formatter

Validators

JSON Validator
HTML Validator
XML Validator - XSD
XPath Tester
Credit Card Number Generator & Validator
Regular Expression Tester
Java Regular Expression Tester
Cron Expression Generator - Quartz

Encoders & Decoders

Url Encoder & Decoder
Base 64 Encoder & Decoder
QR Code Generator

Code Minifiers / Beautifier

JavaScript Beautifier
CSS Beautifier
JavaScript Minifier
CSS Minifier

Converters

XSD Generator
XSLT (XSL Transformer)
XML to JSON Converter
JSON to XML Converter
CSV to XML Converter
CSV to JSON Converter

XPath Tester / Evaluator

Allows you to test your XPath expressions/queries against a XML file. This tool runs better than other existing XPath online tools as it supports most of the XPath functions (string(), number(), name(), string-length() etc.) and does not limit you to working against nodes. It fully supports XPath 2.0 / 3.0 specification. See the [XPath Examples](#) section for details.

The XPath tester fully supports XML namespaces. See the XPath Examples section for details. The namespace prefix "fn" and "math" are reserved to XPath functions.

*The maximum size limit for file upload is 2 megabytes. Results bigger than 500k will be written to a new window for performance reason and to prevent your browser from being unresponsive.

XML Input

Option 1: Copy-paste your XML document here

Option 2: Or upload your XML document

Choose File

No file chosen

UTF-8

XPath expression

Include the XML item type in output:

Yes

TEST XPATH

TEST XPATH IN NEW WINDOW

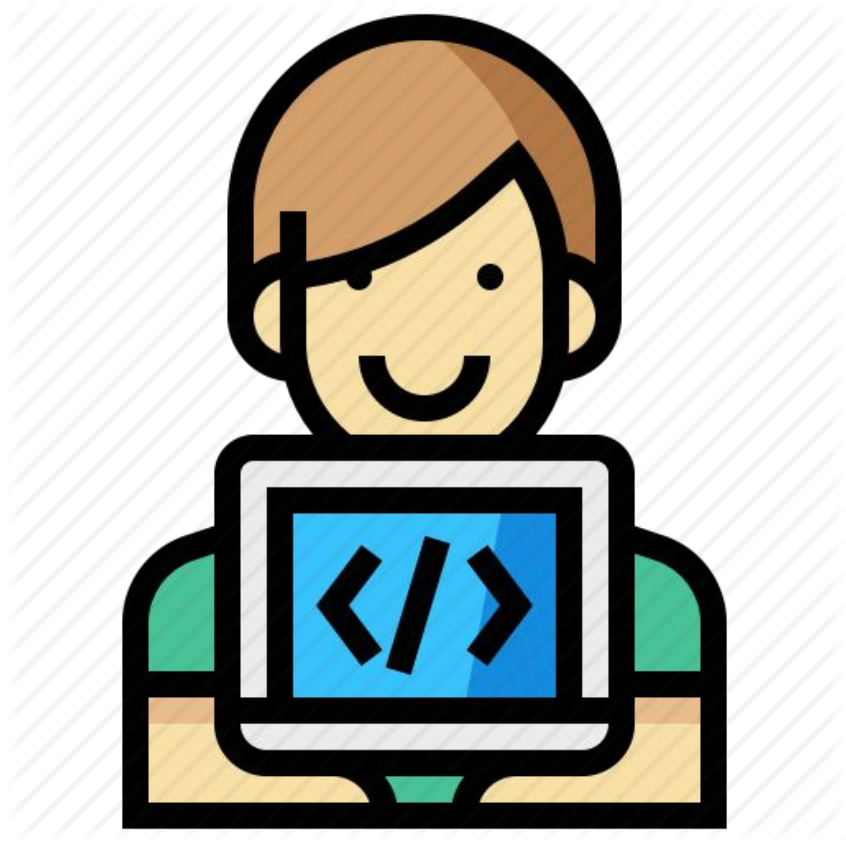
XSLT—XML Transformation

XSLT (Extensible **Stylesheet** Language Transformations) is a language for transforming **XML** documents into other **XML** documents, or other formats such as HTML for web pages, plain text or XSL Formatting Objects, which may subsequently be converted to other formats, such as PDF, PostScript and PNG.

XSLT (extensible Stylesheet Language Transformations) is the recommended style sheet language for XML.

XSLT is far more sophisticated than CSS. With XSLT you can add/remove elements and attributes to or from the output file. You can also rearrange and sort elements, perform tests and make decisions about which elements to hide and display, and a lot more.

XSLT uses XPath to find information in an XML document.



Demonstratio

n

XML file

XSLT working

XSL File

```
C:\Users\ucinf\Documents\uc\17FSIS7024\demo\plants.xml - Notepad++
File Edit Search View Encoding Language Settings Macro Run Plugins Window ?
1 <?xml version="1.0"?>
2 <?xml-stylesheet type="text/xsl" href="plants.xsl"?>
3 <?xml xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
4 xsi:noNamespaceSchemaLocation="plants.xsd"?>
5 <!-- All payload data belongs here -->
6 <genus>Cercis</genus>
7 <species>canadensis</species>
8 <common_name>Eastern Redbud</common_name>
9 <height>25</height>
10 <begin_bloom_date>2017-03-15</begin_bloom_date>
11 <end_bloom_date>2017-03-30</end_bloom_date>
12 <edible>true</edible>
13 <sun_tolerance>full_sun</sun_tolerance>
14 <bloom_color>violet</bloom_color>
15 <specimens>
16   <specimen>
17     <latitude>39.47</latitude>
18     <longitude>-84.51</longitude>
19     <planted_by>Brandan Jones</planted_by>
20     <comments>Beautiful tree!</comments>

```

Notepad++ menu: Edit, Search, View, Encoding, Language, Settings, Tools, Macro, Run, Plugins, Window, ?

product.xml

```
1 <?xml version="1.0"?>
2 <?xml-stylesheet href="http://www.w3.org/1999/XSL/Transform"?>
3 <?xml-stylesheet href="http://www.w3.org/1999/XSL/Transform"?>
4 <?xml-stylesheet href="http://www.w3.org/1999/XSL/Transform"?>
5 <?xml-stylesheet href="http://www.w3.org/1999/XSL/Transform"?>
6 <?xml-stylesheet href="http://www.w3.org/1999/XSL/Transform"?>
7 <?xml-stylesheet href="http://www.w3.org/1999/XSL/Transform"?>
8 <?xml-stylesheet href="http://www.w3.org/1999/XSL/Transform"?>
9 <?xml-stylesheet href="http://www.w3.org/1999/XSL/Transform"?>
10 <?xml-stylesheet href="http://www.w3.org/1999/XSL/Transform"?>
11 <?xml-stylesheet href="http://www.w3.org/1999/XSL/Transform"?>
12 <?xml-stylesheet href="http://www.w3.org/1999/XSL/Transform"?>
13 <?xml-stylesheet href="http://www.w3.org/1999/XSL/Transform"?>
14 <?xml-stylesheet href="http://www.w3.org/1999/XSL/Transform"?>
15 <?xml-stylesheet href="http://www.w3.org/1999/XSL/Transform"?>
16 <?xml-stylesheet href="http://www.w3.org/1999/XSL/Transform"?>
17 <?xml-stylesheet href="http://www.w3.org/1999/XSL/Transform"?>
18 <?xml-stylesheet href="http://www.w3.org/1999/XSL/Transform"?>
19 <?xml-stylesheet href="http://www.w3.org/1999/XSL/Transform"?>
20 <?xml-stylesheet href="http://www.w3.org/1999/XSL/Transform"?>

```

Current Full File path to Clipboard
Current Filename to Clipboard
Current Dir. Path to Clipboard

td>

gth : 599 lines : 24 Ln : 21 Col : 8 Sel : 0 | 0 Windows (CR LF) UTF-8 INS

```
C:\Users\ucinf\Documents\uc\17FSIS7024\demo\plants.xsl - Notepad++
File Edit Search View Encoding Language Settings Macro Run Plugins Window ?
1 <?xml version="1.0" encoding="UTF-8"?>
2 <xsl:stylesheet version="1.0" xmlns:xsl="http://www.w3.org/1999/XSL/Transform">
3 <xsl:template match="/">
4 <html>
5 <head>
6   <meta charset="utf-8" />
7   <title>My Plants</title>
8 </head>
9 <body>
10   <h1>Specimens I have found:</h1>
11   <table style="width:100%;">
12     <tr>
13       <th><xsl:value-of select="latitude"/></th>
14       <th><xsl:value-of select="longitude"/></th>
15       <th><xsl:value-of select="planted_by"/></th>
16       <th><xsl:value-of select="comments"/></th>
17     </tr>
18     <xsl:for-each select="/plant/specimens/specimen">
19       <tr>
20         <td><xsl:value-of select="latitude"/></td>

```

IdeaBoard: Assignments Send Email Calendar Cincinnati My Page file:///C:/Users/ucinf/My Plants My Plants My Plants My Plants

file:///C:/Users/ucinf/17FSIS7024/demo/plants.xsl

Search

Specimens I have found:

39.47	-84.51	Brandan Jones	Beautiful tree!
52.40	1.00	Joe Bloggs	RBG Kew
-22.50	-68.25	Guillermo Sanchez	Lovely Specimen!

Lab Assignment

Q1) Make an XML file about your Favorite Thing, describe its classification like Car □ Toyota □ Honda □ Suzuki □ Nissan etc, then describe more sub-nodes in them and enhance your information by adding attributes.

Q2) Build XSD file to complete the structure and validate it.

Q3) Draw graph of entire XML file by using xmlviewer.

Q4) Code XSL file after considering for XML – HTML conversion.

Q5) Show XSLT file completely(show output after running complete file path to browser).

Q6) Pass 5 different queries by using XPath.